



GW1NZ series of FPGA Products (Automotive) **Datasheet**

DS845-1.1E, 03/16/2023

Copyright © 2023 Guangdong Gowin Semiconductor Corporation. All Rights Reserved.

GOWIN, Gowin, LittleBee, and GOWINSEMI are trademarks of Guangdong Gowin Semiconductor Corporation and are registered in China, the U.S. Patent and Trademark Office, and other countries. All other words and logos identified as trademarks or service marks are the property of their respective holders. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the prior written consent of GOWINSEMI.

Disclaimer

GOWINSEMI assumes no liability and provides no warranty (either expressed or implied) and is not responsible for any damage incurred to your hardware, software, data, or property resulting from usage of the materials or intellectual property except as outlined in the GOWINSEMI Terms and Conditions of Sale. All information in this document should be treated as preliminary. GOWINSEMI may make changes to this document at any time without prior notice. Anyone relying on this documentation should contact GOWINSEMI for the current documentation and errata.

Revision History

Date	Version	Description
03/18/2021	1.0E	Initial version published.
11/25/2021	1.01E	● QN48 info. updated. ● I/O logic info., etc., optimized.
03/16/2023	1.1E	● Note about DC current limit added. ● Section 3.6.4 Byte-enable removed. ● Information on Slew Rate removed. ● Note about the default state of GPIO modified. ● Description of configuration Flash added. ● Information on Slew Rate removed. ● Figure 3-1 GW1NZ-1 Architecture Overview updated. ● Table 3-1 Output I/O Types and Configuration Options of GW1NZ-1 updated. ● Table 4-1 Absolute Max. Ratings updated. ● Table 4-3 Power Supply Ramp Rates updated. ● Table 4-5 POR Specification updated. ● Table 4-8 DC Electrical Characteristics over Recommended Operating Conditions updated. ● Table 4-10 I/O Operating Conditions Recommended updated. ● Table 4-19 User Flash Timing Parameters updated.

Contents

Contents	i
List of Figures	iv
List of Tables	v
1 About This Guide	1
1.1 Purpose	1
1.2 Related Documents	1
1.3 Abbreviations and Terminology	1
1.4 Support and Feedback	2
2 General Description	3
2.1 Features	3
2.2 Product Resources	4
2.3 Package Information	5
3 Architecture	6
3.1 Architecture Overview	6
3.2 Configurable Function Unit	7
3.3 IOB	8
3.3.1 I/O Buffer	9
3.3.2 I/O Logic	11
3.3.3 I/O Logic Modes	14
3.4 I3C Bus (GW1NZ-1)	14
3.4.1 Overview	14
3.4.2 Features	14
3.4.3 Port Signal	15
3.5 SPMI (GW1NZ-1)	16
3.5.1 Overview	16
3.5.2 Port Signal	17
3.6 Block SRAM (BSRAM)	17
3.6.1 Introduction	17
3.6.2 Configuration Mode	18
3.6.3 Mixed Data Bus Width Configuration	19
3.6.4 Parity Bit	20

3.6.5 Synchronous Operation	20
3.6.6 Power up Conditions	20
3.6.7 BSRAM Operation Modes	20
3.6.8 Clock Operations	22
3.7 User Flash (GW1NZ-1).....	23
3.7.1 Introduction	23
3.8 Clock	23
3.8.1 Global Clock	23
3.8.2 PLL	24
3.8.3 HCLK	25
3.9 Long Wire (LW).....	26
3.10 Global Set/Reset (GSR)	26
3.11 Programming Configuration	26
3.11.1 SRAM Configuration	26
3.11.2 Flash Configuration.....	27
3.12 On Chip Oscillator.....	27
4 AC/DC Characteristic	28
4.1 Operating Conditions	28
4.1.1 Absolute Max. Ratings	28
4.1.2 Recommended Operating Conditions	28
4.1.3 Power Supply Ramp Rates	29
4.1.4 Hot Socket Specifications	29
4.1.5 POR Specification.....	29
4.2 ESD	29
4.3 DC Characteristic.....	30
4.3.1 DC Electrical Characteristics over Recommended Operating Conditions	30
4.3.2 Static Current.....	31
4.3.3 I/O Operating Conditions Recommended	32
4.3.4 Single - Ended IO DC Electrical Characteristic.....	33
4.4 Switching Characteristic	34
4.4.1 CFU Block Internal Timing Parameters	34
4.4.2 Gearbox Switching Characteristics.....	34
4.4.3 Clock and I/O Switching Characteristics.....	35
4.4.4 BSRAM Switching Characteristics.....	35
4.4.5 On chip Oscillator Output Frequency	35
4.4.6 PLL Switching Characteristics	36
4.5 User Flash Characteristic	36
4.5.1 DC Characteristic.....	36
4.5.2 Timing Parameters ^{[1],[5],[6]}	37

4.5.3 Operation Timing Diagrams.....	38
4.6 Configuration Interface Timing Specification	39
5 Ordering Information.....	40
5.1 Part Name.....	40
5.2 Package Mark.....	40

List of Figures

Figure 3-1 GW1NZ-1 Architecture Overview	6
Figure 3-2 CFU View.....	8
Figure 3-3 IOB Structure View	9
Figure 3-4 I/O Bank Distribution of GW1NZ-1	9
Figure 3-5 I/O Logic Output	12
Figure 3-6 I/O Logic Input	12
Figure 3-7 IODELAY	13
Figure 3-8 Register Structure in I/O Logic	13
Figure 3-9 IEM Structure.....	14
Figure 3-10 Pipeline Mode in Single Port, Dual Port and Semi Dual Port.....	20
Figure 3-11 Independent Clock Mode.....	22
Figure 3-12 Read/Write Clock Mode.....	22
Figure 3-13 Single Port Clock Mode.....	23
Figure 3-14 PLL Structure.....	24
Figure 3-15 GW1NZ-1 HCLK Distribution.....	26
Figure 4-1 Read Mode	38
Figure 4-2 Write Mode	39
Figure 4-3 Erasure Mode	39
Figure 5-1 Part Naming Example—Production	40
Figure 5-2 Package Mark.....	40

List of Tables

Table 1-1 Abbreviations and Terminologies	2
Table 2-1 Product Resources.....	4
Table 2-2 Package Information and Max. User I/O, True LVDS Pairs	5
Table 3-1 Output I/O Types and Configuration Options of GW1NZ-1.....	10
Table 3-2 Input I/O Types and Configuration Options of GW1NZ-1	11
Table 3-3 Port Description.....	12
Table 3-4 I3C Port Signals	15
Table 3-5 SPMI Port Signal	17
Table 3-6 Memory Size Configuration.....	18
Table 3-7 Dual Port Mixed Read/Write Data Width Configuration	19
Table 3-8 Semi Dual Port Mixed Read/Write Data Width Configuration.....	19
Table 3-9 Clock Operations in Different BSRAM Modes	22
Table 3-10 PLL Ports Definition	25
Table 3-11 Oscillator Output Frequency Options	27
Table 4-1 Absolute Max. Ratings	28
Table 4-2 Recommended Operating Conditions.....	28
Table 4-3 Power Supply Ramp Rates.....	29
Table 4-4 Hot Socket Specifications	29
Table 4-5 POR Specification	29
Table 4-6 GW1NZ ESD - HBM.....	29
Table 4-7 GW1NZ ESD – CDM.....	29
Table 4-8 DC Electrical Characteristics over Recommended Operating Conditions	30
Table 4-9 Static Supply Current (LV Device)	31
Table 4-10 I/O Operating Conditions Recommended	32
Table 4-11 Single - Ended IO DC Electrical Characteristic.....	33
Table 4-12 CFU Block Internal Timing Parameters.....	34
Table 4-13 Gearbox Internal Timing Parameters	34
Table 4-14 Clock and I/O Switching Characteristics	35
Table 4-15 BSRAM Internal Timing Parameters	35
Table 4-16 On chip Oscillator Output Frequency.....	35

Table 4-17 PLL Switching Characteristics.....	36
Table 4-18 GW1NZ-1 User Flash DC Characteristics	36
Table 4-19 User Flash Timing Parameters	37

1 About This Guide

1.1 Purpose

This data sheet describes the features, product resources and structure, AC/DC characteristics, timing specifications of the configuration interface, and the ordering information of the GW1NZ series of FPGA product (Automotive). It is designed to help you understand the GW1NZ series of FPGA product (Automotive) quickly and select and use devices appropriately.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI Website. You can find the related documents at www.gowinsemi.com:

- [UG290, Gowin FPGA Products Programming and Configuration User Guide](#)
- [UG846, GW1NZ series of FPGA products \(Automotive\) Package and Pinout](#)
- [UG845, GW1NZ-1\(Automotive\) Pinout](#)

1.3 Abbreviations and Terminology

The abbreviations and terminology used in this manual are as shown in

Table 1-1 below.

Table 1-1 Abbreviations and Terminologies

Abbreviations and Terminology	Full Name
ALU	Arithmetic Logic Unit
BSRAM	Block Static Random Access Memory
CFU	Configurable Functional Unit
CLS	Configurable Logic Section
CRU	Configurable Routing Unit
DCS	Dynamic Clock Selector
DP	True Dual Port 16K BSRAM
DQCE	Dynamic Quadrant Clock Enable
FPGA	Field Programmable Gate Array
GPIO	Gowin Programmable IO
IOB	Input/output Block
LUT4	Four-input Look-up Table
LUT5	Five-input Look-up Table
LUT6	Six-input Look-up Table
LUT7	Seven-input Look-up Table
LUT8	Eight-input Look-up Table
PLL	Phase Locked Loop
QN48	QFN48
REG	Register
SDP	Semi Dual Port 16K BSRAM
SP	Single Port 16K BSRAM
SPMI	System Power Management Interface

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 General Description

The GW1NZ series of FPGA products (Automotive) are the first generation products in the LittleBee® family. They offer ultra-low power consumption, instant on, low cost, non-volatile, high security, various packages, and flexible usage. They can be widely used in communication, industry control, consumer, video control, etc.

GOWINSEMI provides a new generation of FPGA hardware development environment through market-oriented independent research and development that supports the GW1NZ series of FPGA products (Automotive) and applies to FPGA synthesizing, layout, place and routing, data bitstream generation and download, etc.

2.1 Features

- Zero power consumption
 - 55nm embedded flash technology
 - LV: Supports 1.2V core voltage
 - Clock dynamically turns on and off
 - User Flash dynamically turns on and off
- Power Management Module (GW1NZ-1)
 - SPMI: System power management interface
 - VCC and VCCM are independent in the device
- User Flash (GW1NZ-1)
 - Dynamically turns on and off
 - 64K bits
 - Data Width: 32
 - 10,000 write cycles
 - Greater than ten years' data retention at +85 °C
 - Supports page erasure: 2048 bytes per page
 - Duration: Max. 25ns
 - Electric current
 - a). Read Operation: 2.19 mA/25 ns (V_{CC}) & 0.5 mA/25 ns (V_{CCX}) (MAX);
 - b). Write operation/erase operation: 12/12 mA (MAX)
 - Quick page erasure/Write operation
 - Clock frequency: 40MHz
 - Write operation time: $\leq 16\mu s$

- Page erasure time: ≤ 120 ms
- Configuration Flash (GW1NZ-1)
 - 10,000 write cycles
 - Greater than ten years' data retention at +85 °C
- Multiple I/O Standards
 - GW1NZ-1: LVCMOS33/25/18/15/12; LVTTL33; PCI; LVDS25E, BLVDSE, MLVDSE, LVPECLE, RSDSE
 - Input hysteresis option
 - Supports 4mA, 8mA, 16mA, 24mA, etc. drive options
 - Individual Bus Keeper, Weak Pull-up, Weak Pull-down, and Open Drain option
 - Hot Socket
 - I3C hard core, supports SDR mode
 - Support differential output, rather than differential input
- Abundant Slices
 - Four input LUT (LUT4)
 - Supports shifter register
 - Supports shadow SRAM
- Block SRAM with multiple modes
 - Supports Dual Port, Single Port, and Semi Dual Port
 - Supports bytes write enable
- Flexible PLLs
 - Frequency adjustment (multiplication and division) and phase adjustment
 - Supports global clock
- Built-in Flash programming
 - Instant-on
 - Supports security bit operation
 - Supports AUTO BOOT and DUAL BOOT
- Configuration
 - JTAG configuration
 - Offers up to six GowinCONFIG configuration modes: AUTOBOOT, SSPI, MSPI, CPU, SERIAL, and DUAL BOOT

2.2 Product Resources

Table 2-1 Product Resources

Device	GW1NZ-1
LUT4	1,152
Register	864
Shadow SRAM (bits)	4K
Block SRAM (bits)	72K
PLLs	1
User Flash (bits)	64K
Max. I/O	48
Core Voltage (LV)	1.2V

2.3 Package Information

Table 2-2 Package Information and Max. User I/O, True LVDS Pairs

Package	Pitch (mm)	Size (mm)	GW1NZ-1
QN48	0.4	6 x 6	39

Note!

- In this manual, abbreviations are employed to refer to the package types. See 5.1 Part Name.
- JTAGSEL_N and JTAG pins cannot be used as I/O simultaneously. The Max. User I/O noted in this table is referred to when the four JTAG pins (TCK, TDI, TDO, and TMS) are used as I/O. When mode [2:0] = 001, JTAGSEL_N and the four JTAG pins (TCK, TDI, TDO, and TMS) can be used as GPIO simultaneously, and the Max. user I/O is increased by one.

3 Architecture

3.1 Architecture Overview

Figure 3-1 GW1NZ-1 Architecture Overview

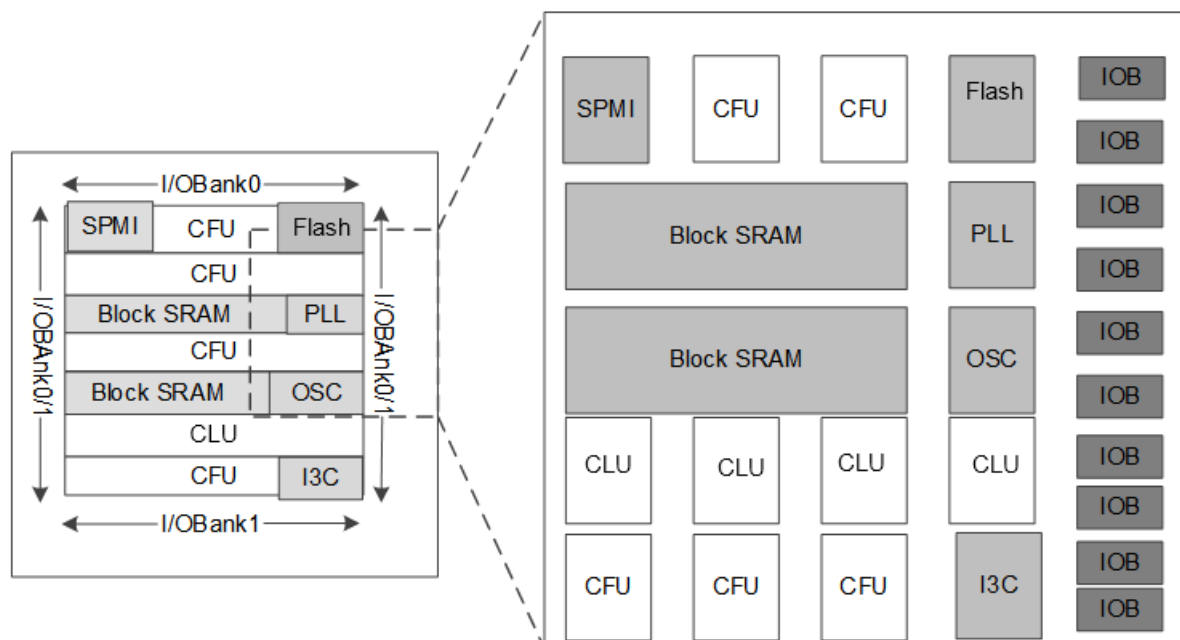


Figure 3-1 shows the GW1NZ devices architecture view. The core of the GW1NZ devices is the array of logic unit surrounded by IO blocks. GW1NZ also provides BSRAM, PLL, Flash, and on chip oscillator and supports Instant-on. SPMI and I3C are also embedded in the GW1NZ devices. See Table 2-1 for more detailed information on internal resources.

Configurable Function Unit (CFU) is the base cell for the array of the GW1NZ series of FPGA products (Automotive). These CFUs arrange in rows and columns. CFU can be configured as LUT4 mode, ALU mode, and memory mode. See [3.2 Configurable Function Unit](#) for further detailed information.

The I/O resources in the GW1NZ series of FPGA products (Automotive) are arranged around the periphery of the devices in groups referred to as banks, including Bank0 and Bank1. The I/O resources

support multiple level standards, and support basic mode, SRD mode, and generic DDR mode. See [3.3 IOB](#) for further detailed information.

The BSRAM is embedded as a row in the GW1NZ series of FPGA products (Automotive). In the FPGA array, each BSRAM occupies three columns of CFU. Each BSRAM has 18,432 bits (18 Kbits) and supports multiple configuration modes and operation modes. See [3.6 Block SRAM \(BSRAM\)](#) for further detailed information.

The GW1N series of FPGA products(Automotive) are embedded with Flash resources, including configuration Flash resources and user Flash resources. Configuration Flash resources are used for internal Flash programming, please refer to [3.11 Programming Configuration](#) for detailed information. User Flash resources are used for user storage, for more detailed information, see [3.7 User Flash \(GW1NZ-1\)](#).

GW1NZ provides one PLL. PLL blocks provide the ability to synthesize clock frequencies. Frequency adjustment (multiplication and division), phase adjustment, and duty cycle can be adjusted using the configuration of parameters. See [3.8 Clock](#) for further detailed information.

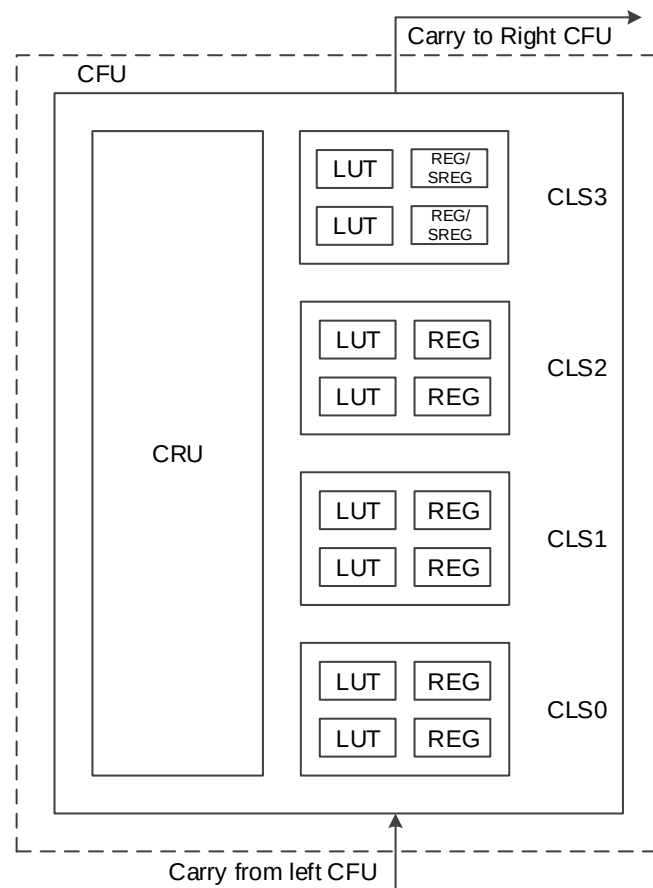
FPGA provides abundant CRUs, connecting all the resources in FPGA. For example, routing resources distributed in CFU and IOB connect resources in CFU and IOB. Routing resources can automatically be generated by Gowin software. In addition, the GW1NZ series of FPGA products (Automotive) also provide abundant GCLKs, long wires (LW), global set/reset (GSR), and programming options, etc. See [3.9Long Wire \(LW\)](#), [3.10Global Set/Reset \(GSR\)](#), [3.11Programming Configuration](#) for further detailed information.

3.2 Configurable Function Unit

The configurable function unit and the configurable logic unit are two basic units for FPGA core of GOWINSEMI. As shown in Figure 3-2, each unit consists of four configurable logic sections and its configurable routing unit. Each of the three configurable logic sections contains two 4-input LUTs and two registers, and the other one only contains two 4-input LUTs.

Configurable logical sections in CLU cannot be configured as SRAM, but as basic logic, ALU, and ROM. The configurable logic sections in the CFU can be configured as basic logic, ALU, SRAM, and ROM depending on the applications.

For further more information about CFU, please refer to [UG288, Gowin Configurable Function Unit \(CFU\) User Guide](#).

Figure 3-2 CFU View**Note!**

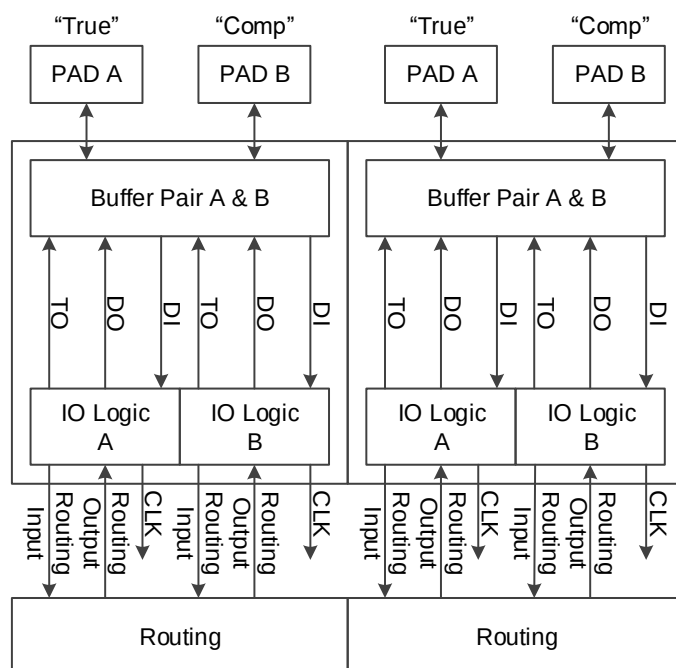
SREG needs special patch support. Please contact Gowin technical support or local Office for this patch.

3.3 IOB

The IOB in the GW1NZ series of FPGA products (Automotive) includes IO buffer, IO logic, and its routing unit. As shown in Figure 3-3, each IOB connects to two Pins (Marked as A and B). As input, they can be used as a single-end signal; as output, they can be used as an output differential pair or as a single end input/output.

Note!

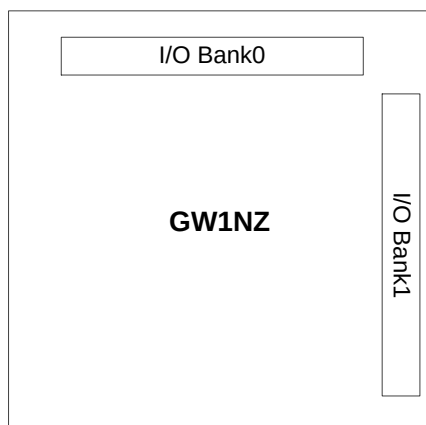
GW1NZ-1 I/Os support differential output, rather than differential input.

Figure 3-3 IOB Structure View**IOB Features:**

- VCCO supplied with each bank
- Supports multiple levels: LVCMOS, PCI, LVTTL, etc.
- Input hysteresis option
- Output drive strength option
- Individual Bus Keeper, Weak Pull-up, Weak Pull-down, and Open Drain option
- Hot Socket
- IO Logic supports basic mode, SRD mode, and generic DDR mode
- I3C hard core embedded, supports SDR mode
- Supports differential output, rather than differential input

3.3.1 I/O Buffer

GW1NZ-1 includes Bank0 and Bank1, as shown in Figure 3-4. Each Bank has independent V_{CCO}. It can be 3.3V, 2.5V, 1.8V, 1.5V, or 1.2V.

Figure 3-4 I/O Bank Distribution of GW1NZ-1

LV devices support 1.2V core voltage to meet users' low power needs. VCCO supplied with I/O Bank can be set as 1.2V, 1.5V, 1.8V, 2.5V, or 3.3V according to requirements. VCCX supports 1.8V, 2.5 V, and 3.3 V power supply.

Note!

During configuration, all GPIOs of the device are internally weak pull-up. After the configuration is complete, the I/O state is controlled by user programs and constraints. The state of CONFIG-related I/Os varies depending on the configuration mode.

For the VCCO requirements of different I/O standards, see Table 3-1 and Table 3-2.

Table 3-1 Output I/O Types and Configuration Options of GW1NZ-1

I/O Type (Output)	Single-ended/Differential	Bank VCCO (V)	Drive Strength (mA)	Typical Applications
LVC MOS33/LVTTL33	Single-ended	3.3	8/24/16/12/4	Universal interface
LVC MOS25	Single-ended	2.5	8/16/12/4	Universal interface
LVC MOS18	Single-ended	1.8	8/12/4	Universal interface
LVC MOS15	Single-ended	1.5	8/4	Universal interface
LVC MOS12	Single-ended	1.2	8/4	Universal interface
PCI33	Single-ended	3.3	8/4	PC and embedded system
LVC MOS33D	Differential	3.3	8/24/16/12/4	Universal interface
LVC MOS25D	Differential	2.5	8/16/12/4	Universal interface
LVC MOS18D	Differential	1.8	8/12/4	Universal interface
LVC MOS15D	Differential	1.5	8/4	Universal interface
LVC MOS12D	Differential	1.2	8/4	Universal interface

Table 3-2 Input I/O Types and Configuration Options of GW1NZ-1

I/O Type (Input)	Single-ended/Differential	Bank $V_{CCO}(V)$	HYSTERESIS	Need V_{REF}
LVC MOS33/LVTT L33	Single-ended	1.2/1.5/1.8/2.5/3.3	Yes	No
LVC MOS25	Single-ended	1.2/1.5/1.8/2.5/3.3	Yes	No
LVC MOS18	Single-ended	1.2/1.5/1.8/2.5/3.3	Yes	No
LVC MOS15	Single-ended	1.2/1.5/1.8/2.5/3.3	Yes	No
LVC MOS12	Single-ended	1.2/1.5/1.8/2.5/3.3	Yes	No
PCI33	Single-ended	3.3	Yes	No
LVC MOS33OD25	Single-ended	2.5	Yes	No
LVC MOS33OD18	Single-ended	1.8	Yes	No
LVC MOS33OD15	Single-ended	1.5	Yes	No
LVC MOS25OD18	Single-ended	1.8	Yes	No
LVC MOS25OD15	Single-ended	1.5	Yes	No
LVC MOS18OD15	Single-ended	1.5	Yes	No
LVC MOS15OD12	Single-ended	1.2	Yes	No
LVC MOS25UD33	Single-ended	3.3	Yes	No
LVC MOS18UD25	Single-ended	2.5	Yes	No
LVC MOS18UD33	Single-ended	3.3	Yes	No
LVC MOS15UD18	Single-ended	1.8	Yes	No
LVC MOS15UD25	Single-ended	2.5	Yes	No
LVC MOS15UD33	Single-ended	3.3	Yes	No
LVC MOS12UD15	Single-ended	1.5	Yes	No
LVC MOS12UD18	Single-ended	1.8	Yes	No
LVC MOS12UD25	Single-ended	2.5	Yes	No
LVC MOS12UD33	Single-ended	3.3	Yes	No

3.3.2 I/O Logic

Figure 3-5 shows the I/O logic output of the GW1NZ series of FPGA

products (Automotive).

Figure 3-5 I/O Logic Output

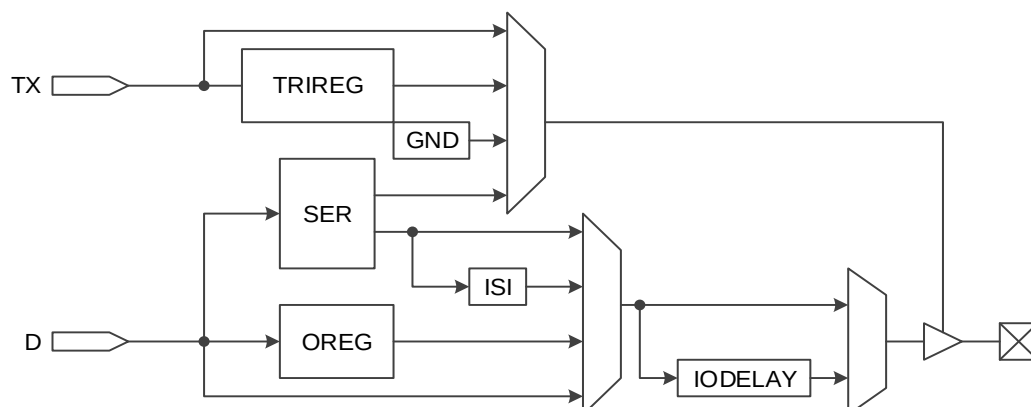


Figure 3-6 shows the I/O logic input of the GW1NZ series of FPGA products (Automotive).

Figure 3-6 I/O Logic Input

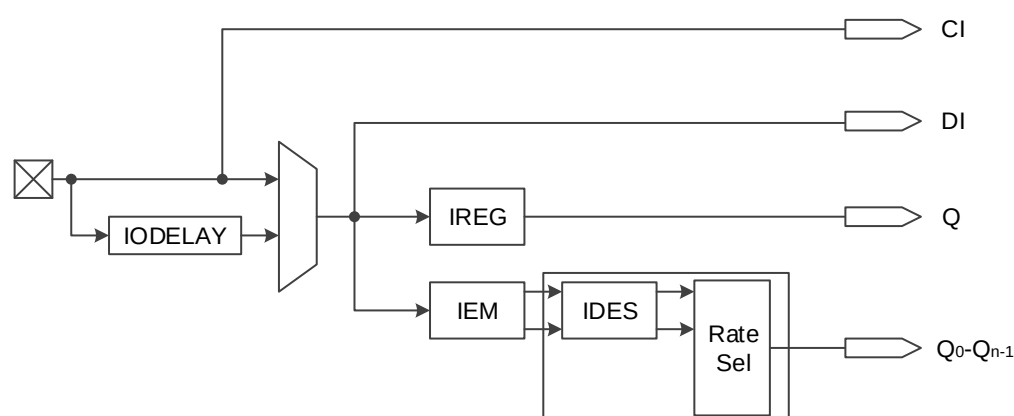


Table 3-3 Port Description

Ports	I/O	Description
C ^[1]	Input	GCLK input signal. For the number of GCLK input signals, please refer to UG845, GW1NZ-1 (Automotive) Pinout .
DI	Input	IO port low-speed input signal, entering into Fabric directly.
Q	Output	IREG output signal in SDR module.
Q ₀ -Q _{n-1}	Output	IDES output signal in DDR module.

Note!

When CI is used as GCLK input, DI, Q, and Q₀-Q_{n-1} cannot be used as I/O input and output.

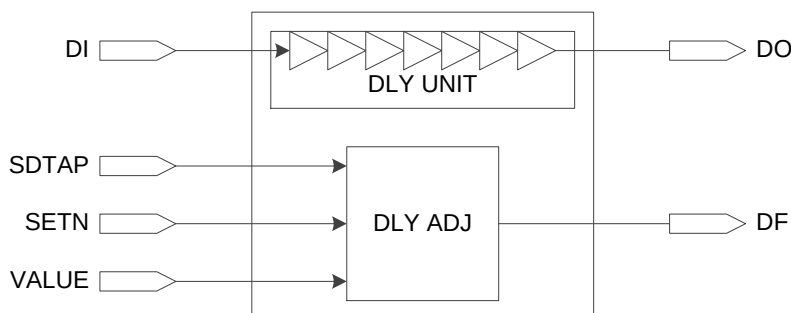
A description of the I/O logic modules of the GW1NZ series of FPGA products (Automotive) is presented below:

IODELAY

See Figure 3-7 for an overview of the IODELAY. Each I/O of the

GW1NZ series of FPGA products (Automotive) has an IODELAY cell. A total of 128(0~127) step delay is provided, with one-step delay time of about 30ps.

Figure 3-7 IODELAY



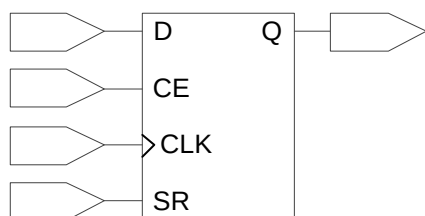
There are two ways to control the delay cell:

- Static control
- Dynamic control: usually used to sample delay window together with IEM. The IODELAY cannot be used for both input and output at the same time

I/O Register

See Figure Figure 3-8 for the I/O register in the GW1NZ series of FPGA products (Automotive). Each I/O provides one input register, INFF, one output register, OUTFF, and a tristate Register, TCFF.

Figure 3-8 Register Structure in I/O Logic

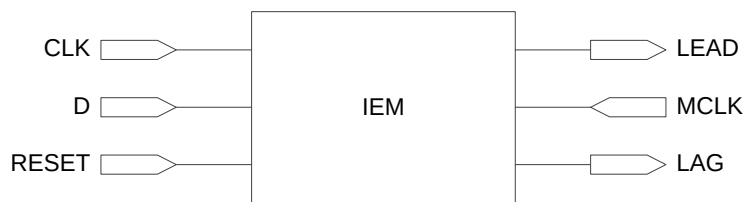


Note!

- CE can be either active low (0: enable) or active high (1: enable).
- CLK can be either rising edge trigger or falling edge trigger.
- SR can be either synchronous/asynchronous SET or RESET or disable.
- The register can be programmed as register or latch.

IEM

IEM is for sampling clock edge and is used in the generic DDR mode, as shown in Figure 3-9.

Figure 3-9 IEM Structure**De-serializer DES and Clock Domain Transfer**

The GW1NZ series of FPGA products (Automotive) provide a simple DES for each input I/O to support advanced I/O protocols.

Serializer SER

The GW1NZ series of FPGA products (Automotive) provide a simple Serializer SER for each output I/O to support advanced I/O protocols.

3.3.3 I/O Logic Modes

The I/O Logic in the GW1NZ series of FPGA products (Automotive) supports several modes. In each operation, the I/O can be configured as output, input, and INOUT or tristate output (output signal with tristate control).

The GW1NZ-1 pins IOR6 (A,B,C....J) do not support IO logic.

For further information about I/O logic modes, please refer to [UG289, Gowin Programmable IO \(GPIO\) User Guide](#).

3.4 I3C Bus (GW1NZ-1)**3.4.1 Overview**

GW1NZ series of FPGA products (Automotive) includes a hard core of I3C bus controller, which supports SDR mode. The I3C controller is backwards compatible with I2C features low power, and is high speed and extensible. The I3C bus is compliant with MIPI I3C protocol, adopts register interfaces, and supports operation modes of I3C SDR Master and I3C SDR Slave.

3.4.2 Features**I3C SDR Master**

- Compliance with MIPI I3C protocol;
- Supports I3C address arbitration detection;
- Supports Single Data Rate (SDR) mode;
- The max. data transmission rate can up to 12.5Mbps;
- Start / Stop / Repeated Start / Acknowledge generation;
- Start / Stop / Repeated Start detection;
- Support dynamically allocating address via SETDASA or ENTDAAs;
- Supports Receive/Send data;
- Supports In-band Interrupts;
- Supports Hot-Join;
- Supports dynamically allocating address when hot-join;

- Supports CCC commands;
- Supports dynamic adjusting SCL frequency;
- Compatible with I2C Slave;
- Adopts register interfaces.

I3C SDR Slave

- Compliance with MIPI I3C protocol;
- Start / Acknowledge generation;
- Start / Stop / Repeated Start detection;
- Support dynamically allocating address via SETDASA or ENTDAAs;
- Receive/Send data;
- Send an IBI or hot-join request. If more than one slaves send the IBI or hot-join requests, the min. address obtains the arbitration;
- Static address of Slave configuration;
- Adopts register interfaces.

3.4.3 Port Signal

For the detailed information about I3C port signals, working principle, timing, and examples, please refer to [IPUG508-1.2 Gowin I3C SDR IP User Guide](#).

Table 3-4 I3C Port Signals

Port Name	I/O	Description
AAC	Input	The setting to clear ACK response, single pulse signal
AAO	output	Output ACK signal
AAS	Input	Set ACK response, single pulse signal
ACC	Input	The setting to clear continuous operation mode, single pulse signal
ACKHS	Input	Set ACK high-level time
ACKLS	Input	Set ACK low-level time
ACO	output	Continuous operation mode output
ACS	Input	Set continuous operation mode, single pulse signal
ADDRS	Input	Set slave address
CE	Input	Clock enable signal
CLK	Input	Clock input
CMC	Input	Clear the current Master role, single pulse signal
CMO	output	Output the flag of current master role
CMS	Input	Set the current Master role, single pulse signal
DI[7:0]	Input	Data input
DO[7:0]	output	Data output
DOBUF[7:0]	output	Buffer data output
LGYC	Input	Clear the setting of I2C as the current communication object, single pulse signal
LGYO	output	Output of I2C as the current communication object

Port Name	I/O	Description
LGYS	Input	Set I2C as the current communication object, single pulse signal
PARITYERROR	output	Parity error signal
RECV DHS	Input	Set high-level time of receiving data
RECV DLS	Input	Set low-level time of receiving data
RESET	Input	Asynchronous reset, active high
SCLI	Input	I3C serial clock input
SCLO	output	I3C serial clock line
SCLOEN	output	I3C serial clock output enable
SCLPULLO	output	I3C serial clock pullup output
SCLPULLOEN	output	I3C serial clock pullup output enable
SDAI	Input	I3C serial data input
SDAO	output	I3C serial data output
SDAOEN	output	I3C serial data output enable
SDAPULLO	output	I3C serial data pullup output
SDAPULLOEN	output	I3C serial data pullup output enable
SENDAHS	Input	Set high-level time of sending address
SENDALS	Input	Set low-level time of sending address
SEND DHS	Input	Set high-level time of sending data
SEND DLS	Input	Set low-level time of sending data
SIC	Input	Signal of clearing interrupt flag
SIO	output	Signal of output signal interrupt
STRTC	Input	Setting of clearing the START command, single pulse signal
STRTO	output	Output START command
STRTS	Input	Set START command, single pulse signal
STATE	output	Output internal state
STRTHDS	Input	Set the holding time of the START command
STOPC	Input	Clear the STOP command setting, single pulse signal
STOPO	output	Output the STOP command
STOPS	Input	Set the STOP command, single pulse signal
STOPSUS	Input	Set the setting time of the STOP command
STOPHDS	Input	Set the holding time of the STOP command

3.5 SPMI (GW1NZ-1)

3.5.1 Overview

The GW1NZ series of FPGA products (Automotive) provides SPMI and the SPMI controller IP. As a Master, the GW1NZ device supports for the power management of the external Slave devices via the SPMI interface. As a Slave, it also supports for the FPGA power management.

The GW1NZ series of FPGA products (Automotive) supports two ways to control the main power:

- Using hardware I/O VCCEN: The main power is turned off when VCCEN is 0. The main power is on when VCCEN is 1;
- Sending the command of shut down by Master: Master sends reset / sleep / wakeup to recover FPGA main power. The main power can also be recovered at low pulse of SPMI_EN.

Note!

For the detailed information of operation modes, communication modes, commands, and timing, etc, please refer to [IPUG529, Gowin SPMI User Guide](#).

3.5.2 Port Signal

Table 3-5 SPMI Port Signal

Name	I/O	Description
SPMI_EN	input	SPMI enable signal
SPMI_CLK	input	System clock signal
SPMI_SCLK	inout	SPMI serial clock signal
SPMI_SDATA	inout	SPMI serial data signal

3.6 Block SRAM (BSRAM)

3.6.1 Introduction

GW1NZ series FPGA products provide abundant SRAM. The Block SRAM (BSRAM) is embedded as a row in the FPGA array and is different from S-SRAM (Shadow SRAM). Each BSRAM occupies three columns of CFU in the FPGA array. Each BSRAM has 18,432 bits (18Kbits). There are five operation modes: Single Port, Dual Port, Semi-dual Port, ROM, and FIFO.

An abundance of BSRAM resources provide a guarantee for the user's high-performance design. BSRAM features:

- Max.18,432 bits per BSRAM
- BSRAM itself can run at 170MHz at max (typical, Read-before-write is 100MHz)
- Single Port
- Dual Port
- Semi-dual Port
- Parity bits
- ROM
- Data width from 1 to 36 bits
- Mixed clock mode
- Mixed data width mode
- Enable Byte operation for double byte or above
- Normal read and write mode
- Read-before-write mode
- Write-through Mode

3.6.2 Configuration Mode

The BSRAM mode in the GW1NZ series of FPGA products (Automotive) supports different data bus widths. See Table 3-6.

Table 3-6 Memory Size Configuration

Single Port Mode	Dual Port Mode ^[1]	Semi-Dual Port Mode
16K x 1	16K x 1	16K x 1
8K x 2	8K x 2	8K x 2
4K x 4	4K x 4	4K x 4
2K x 8	2K x 8	2K x 8
1K x 16	1K x 16	1K x 16
512 x 32	-	512 x 32
2K x 9	2K x 9	2K x 9
1K x 18	1K x 18	1K x 18
512 x 36	-	512 x 36

Note!

[1] The GW1NZ-1 device does not support Dual Port Mode.

Single Port Mode

In the single port mode, BSRAM can write to or read from one port at one clock edge. During the write operation, the data can show up at the output of BSRAM. Normal-Write Mode and Write-through Mode can be supported. When the output register is bypassed, the new data will show at the same write clock rising edge.

For further information about Single Port Block Memory ports and the related description, please refer to [UG285, Gowin BSRAM & SSRAM User Guide](#).

Dual Port Mode

BSRAM support dual port mode. The applicable operations are as follows:

- Two independent read
- Two independent write
- An independent read and an independent write at different clock frequencies

For further information about Dual Port Block Memory ports and the related description, please refer to [UG285, Gowin BSRAM & SSRAM User Guide](#).

Semi-Dual Port Mode

Semi-Dual Port supports read and write at the same time on different ports, but it is not possible to write and read to the same port at the same time. The system only supports write on Port A, read on Port B.

For further information about Semi-Dual Port Block Memory ports and the related description, please refer to [UG285, Gowin BSRAM & SSRAM](#)

[User Guide..](#)

Read Only

BSRAM can be configured as ROM. The ROM can be initialized during the device configuration stage, and the ROM data needs to be provided in the initialization file. Initialization completes during the device power-on process.

Each BSRAM can be configured as one 16 Kbits ROM. For further information about Read Only Port Block Memory ports and the related description, please refer to [UG285, Gowin BSRAM & SSRAM User Guide..](#)

3.6.3 Mixed Data Bus Width Configuration

BSRAM in the GW1NZ series of FPGA products (Automotive) supports mixed data bus width operation. In the dual port and semi-dual port modes, the data bus width for read and write can be different. For the configuration options that are available, please see Table 3-7 and Table 3-8 below.

Table 3-7 Dual Port Mixed Read/Write Data Width Configuration

Read Port	Write Port						
	16K x 1	8K x 2	4K x 4	2K x 8	1K x 16	2K x 9	1K x 18
16K x 1	*	*	*	*	*		
8K x 2	*	*	*	*	*		
4K x 4	*	*	*	*	*		
2K x 8	*	*	*	*	*		
1K x 16	*	*	*	*	*		
2K x 9						*	*
1K x 18						*	*

Note!

"*"denotes the modes supported.

Table 3-8 Semi Dual Port Mixed Read/Write Data Width Configuration

Read Port	Write Port								
	16K x 1	8K x 2	4K x 4	2K x 8	1K x 16	512 x 32	2K x 9	1K x 18	512 x 36
16K x 1	*	*	*	*	*	*			
8K x 2	*	*	*	*	*	*			
4K x 4	*	*	*	*	*	*			
2K x 8	*	*	*	*	*	*			
1K x 16	*	*	*	*	*	*			
512 x 32	*	*	*	*	*	*			
2K x 9							*	*	*
1K x 18							*	*	*

Note!

"*" denotes the modes supported.

3.6.4 Parity Bit

There are parity bits in BSRAMs. The 9th bit in each byte can be used as a parity bit to check the correctness of data transmission. It can also be used for data storage.

3.6.5 Synchronous Operation

- All the input registers of BSRAM support synchronous write;
- The output registers can be used as pipeline register to improve design performance;
- The output registers are bypass-able.

3.6.6 Power up Conditions

BSRAM initialization is supported when powering up. During the power-up process, BSRAM is in standby mode, and all the data outputs are "0". This also applies in ROM mode.

3.6.7 BSRAM Operation Modes

BSRAM supports five different operations, including two read operations (Bypass Mode and Pipeline Read Mode) and three write operations (Normal Write Mode, Write-through Mode, and Read-before-write Mode).

Read Mode

Read data from the BSRAM via output registers or without using the registers.

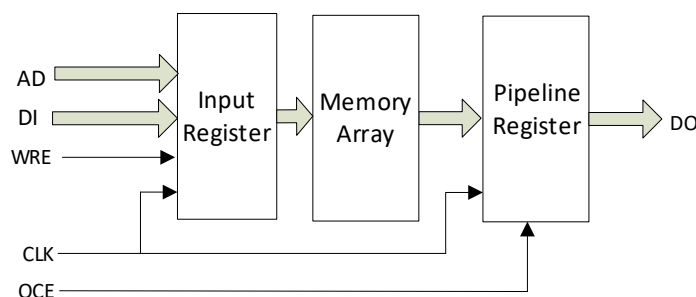
Pipeline Mode

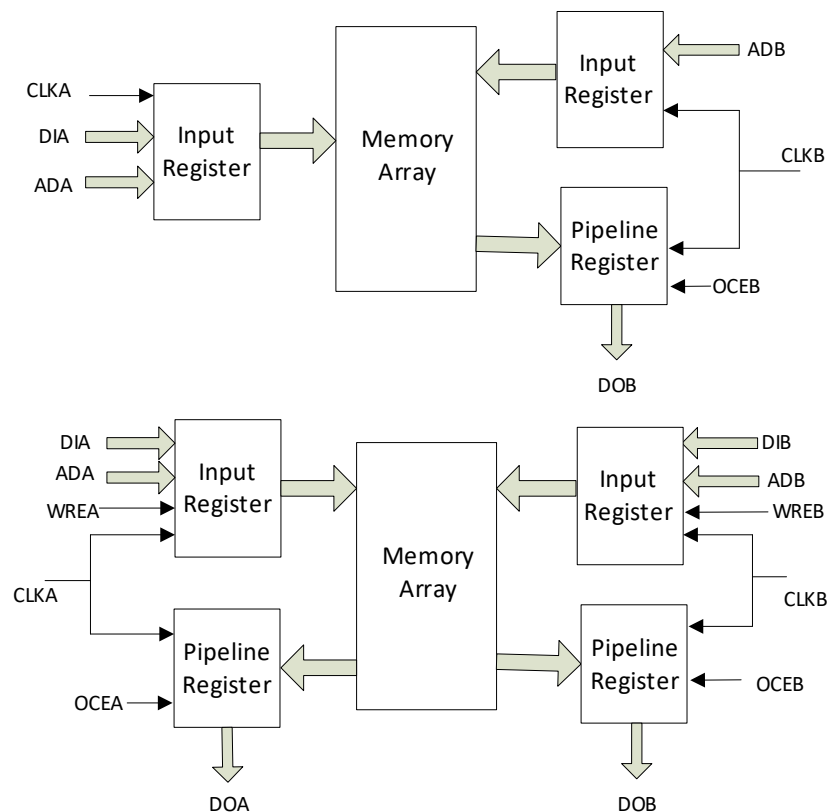
While writing in the BSRAM, the output register and pipeline register are also being written. The data bus can be up to 36 bits in this mode.

Bypass Mode

The output register is not used. The data is kept in the output of memory array.

Figure 3-10 Pipeline Mode in Single Port, Dual Port and Semi Dual Port





Write Mode

NORMAL WRITE MODE

In this mode, when the user writes data to one port, and the output data of this port does not change. The data written in will not appear at the read port.

WRITE-THROUGH MODE

In this mode, when the user writes data to one port, and the data written in will also appear at the output of this port.

READ-BEFORE-WRITE MODE

In this mode, when the user writes data to one port, and the data written in will be stored in the memory according to the address. The original data in this address will appear at the output of this port.

3.6.8 Clock Operations

Table 3-9 lists the clock operations in different BSRAM modes:

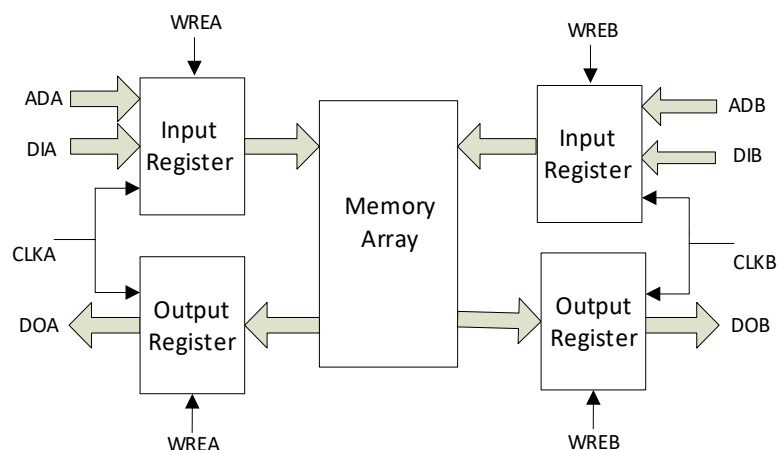
Table 3-9 Clock Operations in Different BSRAM Modes

Clock Operations	Dual Port Mode	Semi-Dual Port Mode	Single Port Mode
Independent Clock Mode	Yes	No	No
Read/Write Clock Mode	Yes	Yes	No
Single Port Clock Mode	No	No	Yes

Independent Clock Mode

Figure 3-11 shows the independent clocks in dual port mode with each port with one clock. CLKA controls all the registers at Port A; CLKB controls all the registers at Port B.

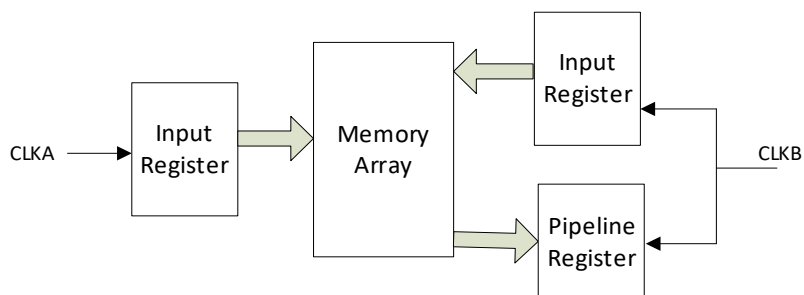
Figure 3-11 Independent Clock Mode



Read/Write Clock Operation

Figure 3-12 shows the read/write clock operations in the semi-dual port mode with one clock at each port. The write clock (CLKA) controls Port A data inputs, write address and write enable signals. The read clock (CLKB) controls Port B data output, read address, and read enable signals.

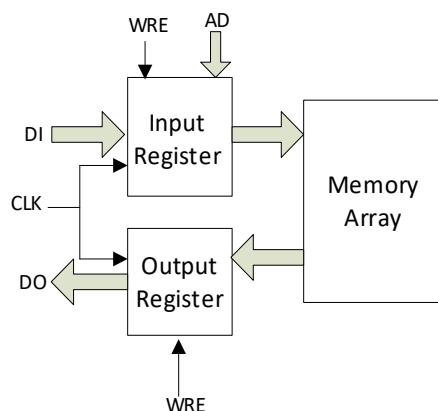
Figure 3-12 Read/Write Clock Mode



Single Port Clock Mode

Figure 3-13 shows the clock operation in single port mode.

Figure 3-13 Single Port Clock Mode



3.7 User Flash (GW1NZ-1)

3.7.1 Introduction

GW1NZ-1 offers User Flash, the features are shown below:

- 10,000 write cycles
- Capacity: 64K bits
- Greater than ten years' data retention at +85 °C
- Supports page erasure: 2,048 bytes per page
- Quick page erasure/Write operation
- Clock frequency: 40MHz
- Write operation time: ≤16μs
- Page erasure time: ≤120 ms
- Electric current
 - Read Operation: 2.19 mA/25 ns (V_{CC}) & 0.5 mA/25 ns (V_{CCX}) (MAX).
 - Write operation/erase operation: 12/12 mA(MAX)

For further information about the user Flash in GW1NZ-1, please refer to [UG295, Gowin User Flash User Guide](#). For the correspondence between user Flash primitives and devices supported, please refer to Table 3-1 Devices Supported of [UG295, Gowin User Flash User Guide](#).

3.8 Clock

For further detailed information, please refer to [UG286, Gowin Clock User Guide](#).

3.8.1 Global Clock

The GCLK is distributed in GW1NZ-1 as two quadrants, L and R. Each quadrant provides eight GCLKs. Each GCLK has 12 optional clock sources. The optional clock resources of GCLK can be pins or CRU. Users can employ dedicated pins as clock resources to achieve better

timing.

3.8.2 PLL

GW1NZ-1

PLL (Phase-locked Loop) is one kind of a feedback control circuit. The frequency and phase of the internal oscillator signal is controlled by the external input reference clock.

PLL blocks provide the ability to synthesize clock frequencies. Frequency adjustment (multiplication and division), phase adjustment, and duty cycle can be adjusted by configuring the parameters.

See Figure 3-14 for the PLL structure.

Figure 3-14 PLL Structure

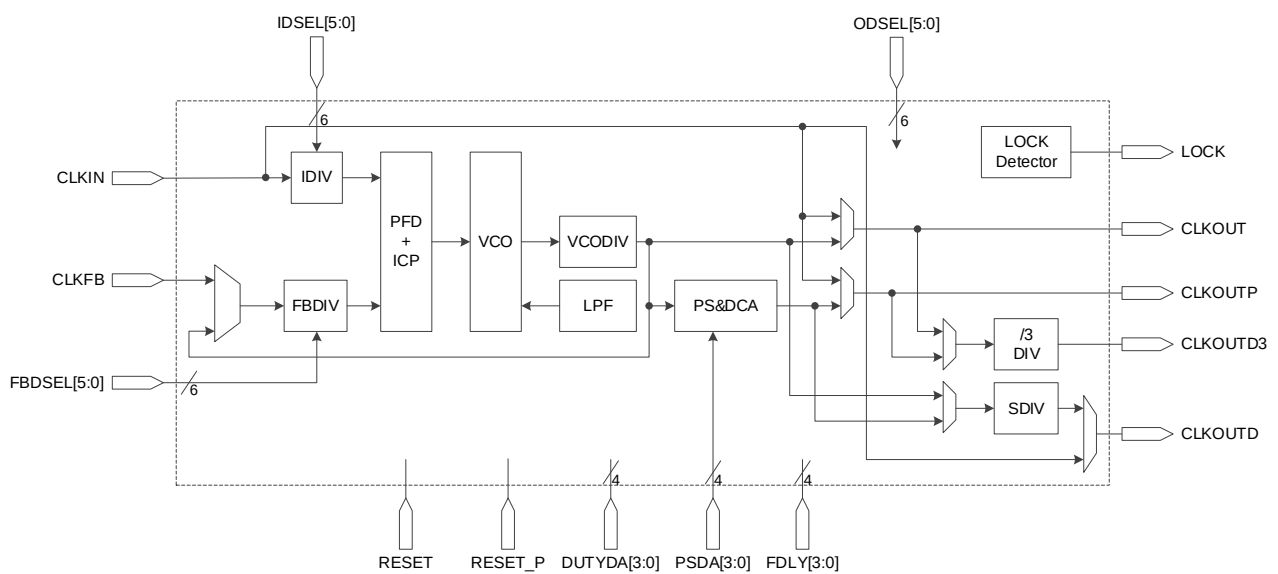


Table 3-10 PLL Ports Definition

Port Name	Signal	Description
CLKIN[5:0]	I	Reference clock input
CLKFB	I	Feedback clock input
RESET	I	PLL reset
RESET_P	I	PLL Power Down
IDSEL [5:0]	I	Dynamic IDIV control: 1~64
FBDSEL [5:0]	I	Dynamic FBDIV control: 1~64
PSDA [3:0]	I	Dynamic phase control (rising edge effective)
DUTYDA [3:0]	I	Dynamic duty cycle control (falling edge effective)
FDLY[3:0]	I	CLKOUTP dynamic delay control
CLKOUT	Output	Clock output with no phase and duty cycle adjustment
CLKOUTP	Output	Clock output with phase and duty cycle adjustment
CLKOUTD	Output	Clock divider from CLKOUT and CLKOUTP (controlled by SDIV)
CLKOUTD3	Output	clock divider from CLKOUT and CLKOUTP (controlled by DIV3 with the constant division value 3)
LOCK	Output	PLL lock status: 1 locked, 0 unlocked

The PLL reference clock source can come from an external PLL pin or from internal routing GCLK, HCLK, or general data signal. PLL feedback signal can come from the external PLL feedback input or from internal routing GCLK, HCLK, or general data signal.

For PLL features of GW1NZ series of FPGA products (Automotive), please refer to [4.4.6 PLL Switching Characteristics](#).

PLL can adjust the frequency of the input clock CLKIN (multiplication and division). The formulas for doing so are as follows:

- $f_{CLKOUT} = (f_{CLKIN} * FBDIV) / IDIV$;
- $f_{VCO} = f_{CLKOUT} * ODIV$;
- $f_{CLKOUTD} = f_{CLKOUT} / SDIV$;
- $f_{PFD} = f_{CLKIN} / IDIV = f_{CLKOUT} / FBDIV$.

Note!

- f_{CLKIN} : The frequency of input clock CLKIN;
- f_{CLKOUT} : The clock frequency of CLKOUT and CLKOUTP
- $f_{CLKOUTD}$: The clock frequency of CLKOUTD, and CLKOUTD is the clock CLKOUT after division
- f_{PFD} : PFD Frequency, and the minimum value of f_{PFD} should be no less than 3MHz.

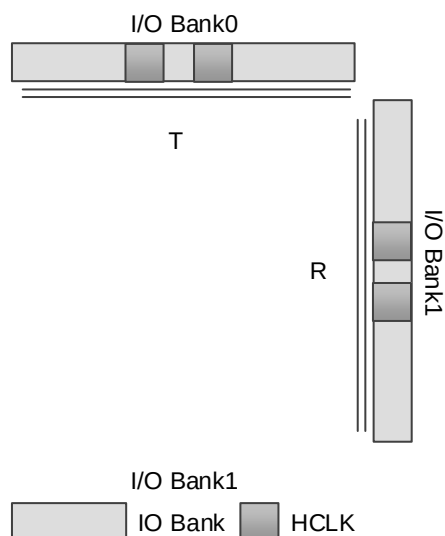
Adjust IDIV, FBDIV, ODIV, and SDIV to achieve the required clock frequency.

3.8.3 HCLK

HCLK is the high-speed clock in the GW1NZ series of FPGA products (Automotive). It can support high-performance data transfer and is mainly suitable for source synchronous data transfer protocols. See Figure Figure

3-15. HCLK can be used for the whole I/O Bank.

Figure 3-15 GW1NZ-1 HCLK Distribution



3.9 Long Wire (LW)

As a supplement to the CRU, the GW1NZ series of FPGA products (Automotive) provides another routing resource, Long wire, which can be used as clock, clock enable, set/reset, or other high fan out signals.

3.10 Global Set/Reset (GSR)

A global set/reset (GSR) network is built in the GW1NZ series of FPGA product (Automotive). There is a direct connection to core logic. It can be used as asynchronous/synchronous set. The registers in CFU and I/O can be individually configured to use GSR.

3.11 Programming Configuration

The GW1NZ series of FPGA products (Automotive) support SRAM and Flash. The Flash programming mode supports on-chip Flash and off-chip Flash. The GW1NZ series of FPGA products (Automotive) support DUAL BOOT, providing a selection for users to backup data to off chip Flash according to requirements.

Besides JTAG, the GW1NZ series of FPGA products (Automotive) also supports GOWINSEMI's own configuration mode: GowinCONFIG (AUTO BOOT, SSPI, MSPI, DUAL BOOT, SERIAL, CPU, and I²C Slave). All the devices support JTAG and AUTO BOOT. For the detailed information, please refer to [UG290, Gowin FPGA Products Programming and Configuration User Guide](#).

3.11.1 SRAM Configuration

When you adopt SRAM to configure the device, every time the device is powered on, the bit stream file needs to be downloaded to configure the device.

3.11.2 Flash Configuration

The Flash configuration data is stored in the on-chip flash. Each time the device is powered on, the configuration data is transferred from the Flash to the SRAM, which controls the working of the device. This mode can complete configuration within a few ms, and is referred to as “Quick Start”. The GW1NZ series of FPGA products (Automotive) also support off-chip Flash configuration and dual-boot. Please refer to [UG290, Gowin FPGA Products Programming and Configuration User Guide](#) for more detailed information.

3.12 On Chip Oscillator

There is an on chip oscillator in each of the GW1NZ series of FPGA product (Automotive). The on chip oscillator provides a programmable user clock with precision of $\pm 5\%$. During the configuration process, it can provide a clock for the MSPI mode. See Table 3-11 for the output frequency.

Table 3-11 Oscillator Output Frequency Options

Mode	Frequency	Mode	Frequency	Mode	Frequency
0	2.5MHz ^[1]	8	7.8MHz	16	15.6MHz
1	5.4MHz	9	8.3MHz	17	17.9MHz
2	5.7MHz	10	8.9MHz	18	21MHz
3	6.0MHz	11	9.6MHz	19	25MHz
4	6.3MHz	12	10.4MHz	20	31.3MHz
5	6.6MHz	13	11.4MHz	21	41.7MHz
6	6.9MHz	14	12.5MHz	22	62.5MHz
7	7.4MHz	15	13.9MHz	23	125MHz ^[2]

Note!

- [1] The default frequency is 2.5MHz.
- [2] 125 MHz is not suitable for MSPI.

The on-chip oscillator also provides a clock resource for user designs. Up to 64 clock frequencies can be obtained by setting the parameters. The following formula is employed to get the output clock frequency:

$$f_{\text{out}} = 250\text{MHz} / \text{Param.}$$

“Param” is the configuration parameter with a range of 2~128. It supports even number only.

4 AC/DC Characteristic

4.1 Operating Conditions

4.1.1 Absolute Max. Ratings

Table 4-1 Absolute Max. Ratings

Name	Description	Min.	Max.
V _{CC}	Core voltage	-0.5V	1.32V
V _{CCO}	I/O Bank Power	-0.5V	3.75V
V _{CCX}	Auxiliary Power	-0.5V	3.75V
-	I/O Voltage Applied ^[1]	-0.5V	3.75V
Storage Temperature	Storage Temperature	-65°C	+150°C
Junction Temperature	Junction Temperature	-40°C	+125°C

Note!

[1] Overshoot and undershoot of -2V to (V_{IHMAX} + 2)V are allowed for a duration of <20 ns.

4.1.2 Recommended Operating Conditions

Table 4-2 Recommended Operating Conditions

Name	Description	Min.	Max.
V _{CC}	LV: Core Power	1.14V	1.26V
V _{CCO}	I/O Bank Power	1.14V	3.6V
V _{CCX}	Auxiliary voltage	1.71V	3.6V
T _{JAUT}	Junction temperature Automotive operation	-40°C	+105°C

Note!

For the detailed recommended operating conditions for different packages, please refer to [UG845, GW1NZ-1 \(Automotive\) Pinout](#).

4.1.3 Power Supply Ramp Rates

Table 4-3 Power Supply Ramp Rates

Name	Description	Min.	Typ.	Max.
V _{CC} Ramp	Power supply ramp rates for V _{CC}	1.2mV/μs	-	40mV/μs
V _{CCX} Ramp	Power supply ramp rates for V _{CCX}	0.6mV/μs	-	10mV/us
V _{CCIO} Ramp	Power supply ramp rates for V _{CCIO}	0.1mV/μs	-	10mV/us

4.1.4 Hot Socket Specifications

Table 4-4 Hot Socket Specifications

Name	Description	Condition	I/O Type	Max.
I _{HS}	Input or I/O leakage current	0<V _{IN} <V _{IH} (MAX)	I/O	150uA
I _{HS}	Input or I/O leakage current	0<V _{IN} <V _{IH} (MAX)	TDI,TDO, TMS,TCK	120uA

4.1.5 POR Specification

Table 4-5 POR Specification

Name	Description	Device	Name	Value
V _{POR_UP}	Power on reset ramp up trip point	GW1NZ-1	V _{CC}	0.8V
			V _{CCX}	1.5V
			V _{CCIO}	0.9V
V _{POR_DOWN}	Power on reset ramp down trip point	GW1NZ-1	V _{CC}	0.65V
			V _{CCX}	1.4V
			V _{CCIO}	0.7V

4.2 ESD

Table 4-6 GW1NZ ESD - HBM

Device	GW1NZ-1
QN48	HBM>1,000V

Table 4-7 GW1NZ ESD - CDM

Device	GW1NZ-1
QN48	CDM>500V

4.3 DC Characteristic

4.3.1 DC Electrical Characteristics over Recommended Operating Conditions

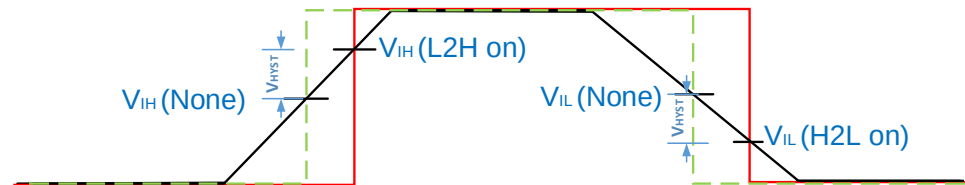
Table 4-8 DC Electrical Characteristics over Recommended Operating Conditions

Name	Description	Condition	Min.	Typ.	Max.
I_{IL}, I_{IH}	Input or I/O leakage	$V_{CCO} < V_{IN} < V_{IH} \text{ (MAX)}$	-	-	210 μA
		$0V < V_{IN} < V_{CCO}$	-	-	10 μA
I_{PU}	I/O Active Pull-up Current	$0 < V_{IN} < 0.7V_{CCO}$	-30 μA	-	-150 μA
I_{PD}	I/O Active Pull-down Current	$V_{IL} \text{ (MAX)} < V_{IN} < V_{CCO}$	30 μA	-	150 μA
I_{BHLS}	Bus Hold Low Sustaining Current	$V_{IN} = V_{IL} \text{ (MAX)}$	30 μA	-	-
I_{BHHS}	Bus Hold High Sustaining Current	$V_{IN} = 0.7V_{CCO}$	-30 μA	-	-
I_{BHL0}	Bus Hold Low Overdrive Current	$0 \leq V_{IN} \leq V_{CCO}$	-	-	150 μA
I_{BHH0}	Bus Hold High Overdrive Current	$0 \leq V_{IN} \leq V_{CCO}$	-	-	-150 μA
V_{BHT}	Bus hold trigger points		$V_{IL} \text{ (MAX)}$	-	$V_{IH} \text{ (MIN)}$
C1	I/O Capacitance			5 pF	8 pF
V_{HYST}	Hysteresis for Schmitt Trigger inputs	$V_{CCIO} = 3.3V, \text{ Hysteresis} = L2H^{[1],[2]}$	-	200mV	-
		$V_{CCIO} = 2.5V, \text{ Hysteresis} = L2H$	-	125mV	-
		$V_{CCIO} = 1.8V, \text{ Hysteresis} = L2H$	-	60mV	-
		$V_{CCIO} = 1.5V, \text{ Hysteresis} = L2H$	-	40mV	-
		$V_{CCIO} = 1.2V, \text{ Hysteresis} = L2H$	-	20mV	-
		$V_{CCIO} = 3.3V, \text{ Hysteresis} = H2L^{[1],[2]}$	-	200mV	-
		$V_{CCIO} = 2.5V, \text{ Hysteresis} = H2L$	-	125mV	-
		$V_{CCIO} = 1.8V, \text{ Hysteresis} = H2L$	-	60mV	-
		$V_{CCIO} = 1.5V, \text{ Hysteresis} = H2L$	-	40mV	-
		$V_{CCIO} = 1.2V, \text{ Hysteresis} = H2L$	-	20mV	-
		$V_{CCIO} = 3.3V, \text{ Hysteresis} = HIGH^{[1],[2]}$	-	400mV	-
		$V_{CCIO} = 2.5V, \text{ Hysteresis} = HIGH$	-	250mV	-
		$V_{CCIO} = 1.8V, \text{ Hysteresis} = HIGH$	-	120mV	-
		$V_{CCIO} = 1.5V, \text{ Hysteresis} = HIGH$	-	80mV	-
		$V_{CCIO} = 1.2V, \text{ Hysteresis} = HIGH$	-	40mV	-

Note!

- [1] Hysteresis="NONE", "L2H", "H2L", "HIGH" indicates the Hysteresis options that can be set when setting I/O Constraints in the FloorPlanner tool of Gowin EDA, for more details, see [SUG935, Gowin Design Physical Constraints User Guide](#).

- [2] Enabling the L2H (low to high) option means raising V_{IH} by V_{HYST} ; enabling the H2L (high to low) option means lowering V_{IL} by V_{HYST} ; enabling the HIGH option means enabling both L2H and H2L options, i.e. $V_{HYST}(HIGH) = V_{HYST}(L2H) + V_{HYST}(L2H)$. The diagram is shown below.



4.3.2 Static Current

Table 4-9 Static Supply Current (LV Device)

Name	Description	Device	Typ.
I_{CC}	Core current ($V_{CC}=1.2V$)	GW1NZ-1	3mA
I_{CCX}	V_{CCX} current ($V_{CCX}=3.3V$)	GW1NZ-1	-
	V_{CCX} current ($V_{CCX}=2.5V$)	GW1NZ-1	-
I_{CCO}	I/O Bank current ($V_{CCO}=2.5V$)	GW1NZ-1	-

Note!

- After device wake up, user can turn off external V_{CCX} when they do not use user flash and chip still functional.
- The typical values in the table above are tested at room temperature.
- In zero power circumstance, if users use MODE pin, the PULL_MODE of this pin must be configured as KEEPER.

4.3.3 I/O Operating Conditions Recommended

Table 4-10 I/O Operating Conditions Recommended

Name	Output V_{CCO} (V)			Input V_{REF} (V)		
	Min.	Typ.	Max.	Min.	Typ.	Max.
LVTTL33	3.135	3.3	3.6	-	-	-
LVC MOS33	3.135	3.3	3.6	-	-	-
LVC MOS25	2.375	2.5	2.625	-	-	-
LVC MOS18	1.71	1.8	1.89	-	-	-
LVC MOS15	1.425	1.5	1.575	-	-	-
LVC MOS12	1.14	1.2	1.26	-	-	-
SSTL15	1.425	1.5	1.575	0.68	0.75	0.9
SSTL18_I	1.71	1.8	1.89	0.833	0.9	0.969
SSTL18_II	1.71	1.8	1.89	0.833	0.9	0.969
SSTL25_I	2.375	2.5	2.645	1.15	1.25	1.35
SSTL25_II	2.375	2.5	2.645	1.15	1.25	1.35
SSTL33_I	3.135	3.3	3.6	1.3	1.5	1.7
SSTL33_II	3.135	3.3	3.6	1.3	1.5	1
HSTL18_I	1.71	1.8	1.89	0.816	0.9	1.08
HSTL18_II	1.71	1.8	1.89	0.816	0.9	1.08
HSTL15	1.425	1.5	1.575	0.68	0.75	0.9
PCI33	3.135	3.3	3.6	-	-	-
LVPECL33E	3.135	3.3	3.6	-	-	-
MLVDS25E	2.375	2.5	2.625	-	-	-
BLVDS25E	2.375	2.5	2.625	-	-	-
RSDS25E	2.375	2.5	2.625	-	-	-
LVDS25E	2.375	2.5	2.625	-	-	-
SSTL15D	1.425	1.5	1.575	-	-	-
SSTL18D_I	1.71	1.8	1.89	-	-	-
SSTL18D_II	1.71	1.8	1.89	-	-	-
SSTL25D_I	2.375	2.5	2.625	-	-	-
SSTL25D_II	2.375	2.5	2.625	-	-	-
SSTL33D_I	3.135	3.3	3.6	-	-	-
SSTL33D_II	3.135	3.3	3.6	-	-	-
HSTL15D	1.425	1.575	1.89	-	-	-
HSTL18D_I	1.71	1.8	1.89	-	-	-
HSTL18D_II	1.71	1.8	1.89	-	-	-

4.3.4 Single - Ended IO DC Electrical Characteristic

Table 4-11 Single - Ended IO DC Electrical Characteristic

Name	V _{IL}		V _{IH}		V _{OL} (Max)	V _{OH} (Min)	I _{OL} ^[1] (mA)	I _{OH} ^[1] (mA)
	Min	Max	Min	Max				
LVCMOS33 LVTTTL33	-0.3V	0.8V	2.0V	3.6V	0.4V	V _{CCO} -0.4V	4	-4
							8	-8
							12	-12
							16	-16
					0.2V	V _{CCO} -0.2V	0.1	-0.1
LVCMOS25	-0.3V	0.7V	1.7V	3.6V	0.4V	V _{CCO} -0.4V	4	-4
							8	-8
							12	-12
					0.2V	V _{CCO} -0.2V	0.1	-0.1
LVCMOS18	-0.3V	0.35 x V _{CCO}	0.65 x V _{CCO}	3.6V	0.4V	V _{CCO} 0.4V	4	-4
							8	-8
							12	-12
					0.2V	V _{CCO} -0.2V	0.1	-0.1
LVCMOS15	-0.3V	0.35 x V _{CCO}	0.65 x V _{CCO}	3.6V	0.4V	V _{CCO} -0.4V	4	-4
					0.2V	V _{CCO} -0.2V	0.1	-0.1
LVCMOS12	-0.3V	0.35 x V _{CCO}	0.65 x V _{CCO}	3.6V	0.4V	V _{CCO} -0.4V	2	-2
					0.2V	V _{CCO} -0.2V	0.1	-0.1
PCI33	-0.3V	0.3 x V _{CCO}	0.5 x V _{CCO}	3.6V	0.1 V _{CCO} x	0.9 x V _{CCO}	1.5	-0.5
SSTL33_I	-0.3V	V _{REF} -0.2V	V _{REF} +0.2V	3.6V	0.7	V _{CCO} -1.1V	8	-8
SSTL25_I	-0.3V	V _{REF} -0.18V	V _{REF} +0.18V	3.6V	0.54V	V _{CCO} -0.62V	8	-8
SSTL25_II	-0.3V	V _{REF} -0.18V	V _{REF} +0.18V	3.6V	N/A	N/A	N/A	N/A
SSTL18_II	-0.3V	V _{REF} -0.125V	V _{REF} +0.125V	3.6V	N/A	N/A	N/A	N/A
SSTL18_I	-0.3V	V _{REF} -0.125V	V _{REF} +0.125V	3.6V	0.40V	V _{CCO} -0.40V	8	-8
SSTL15	-0.3V	V _{REF} -0.1V	V _{REF} + 0.1V	3.6V	0.40V	V _{CCO} -0.40V	8	-8
HSTL18_I	-0.3V	V _{REF} -0.1V	V _{REF} + 0.1V	3.6V	0.40V	V _{CCO} -0.40V	8	-8
HSTL18_II	-0.3V	V _{REF} -0.1V	V _{REF} + 0.1V	3.6V	N/A	N/A	N/A	N/A
HSTL15_I	-0.3V	V _{REF} -0.1V	V _{REF} + 0.1V	3.6V	0.40V	V _{CCO} -0.40V	8	-8
HSTL15_II	-0.3V	V _{REF} -0.1V	V _{REF} + 0.1V	3.6V	N/A	N/A	N/A	N/A

Note!

[1] The total DC current limit(sourced and sinked) of all IOs in the same bank: the total DC current of all IOs in the same bank shall not be greater than $n \cdot 8\text{mA}$, where n represents the number of IOs bonded out from a bank.

4.4 Switching Characteristic

4.4.1 CFU Block Internal Timing Parameters

Table 4-12 CFU Block Internal Timing Parameters

Name	Description	Speed Grade		Unit
		Min	Max	
$t_{\text{LUT4_CFU}}$	LUT4 delay	-	0.674	ns
$t_{\text{LUT5_CFU}}$	LUT5 delay	-	1.388	ns
$t_{\text{LUT6_CFU}}$	LUT6 delay	-	2.01	ns
$t_{\text{LUT7_CFU}}$	LUT7 delay	-	2.632	ns
$t_{\text{LUT8_CFU}}$	LUT8 delay	-	3.254	ns
$t_{\text{SR_CFU}}$	Set/Reset to Register output	-	1.86	ns
$t_{\text{CO_CFU}}$	Clock to Register output	-	0.76	ns

4.4.2 Gearbox Switching Characteristics

Table 4-13 Gearbox Internal Timing Parameters

TBD

4.4.3 Clock and I/O Switching Characteristics

Table 4-14 Clock and I/O Switching Characteristics

Name	Description	Device	-5		-6		Unit
			Min	Max	Min	Max	
Clocks	TBD	TBD	TBD	TBD	TBD	TBD	
Pin-LUT-Pin Delay	TBD	TBD	TBD	TBD	TBD	TBD	
General I/O Pin Parameters	TBD	TBD	TBD	TBD	TBD	TBD	

4.4.4 BSRAM Switching Characteristics

Table 4-15 BSRAM Internal Timing Parameters

Name	Description	Speed Grade		Unit
		Min	Max	
t _{COAD} _BSRAM	Clock to output from read address/data	-	5.10	ns
t _{COOR} _BSRAM	Clock to output from output register	-	0.56	ns

4.4.5 On chip Oscillator Output Frequency

Table 4-16 On chip Oscillator Output Frequency

Name	Description	Min.	Typ.	Max.
f _{MAX}	Output Frequency(-40 to +105°C)	TBD	TBD	TBD
t _{DT}	Output Clock Duty Cycle	43%	50%	57%
t _{OPJIT}	Output Clock Period Jitter	0.01UIPP	0.012UIPP	0.02UIPP

4.4.6 PLL Switching Characteristics

Table 4-17 PLL Switching Characteristics

Device		Speed Grade	Name	Min.	Max.
GW1NZ-1	LV	A3	CLKIN	3MHz	320MHz
			PFD	3MHz	320MHz
			VCO	320MHz	640MHz
			CLKOUT	2.5MHz	360MHz

4.5 User Flash Characteristic

4.5.1 DC Characteristic

Table 4-18 GW1NZ-1 User Flash DC Characteristics

Name	Parameter	Max.		Unit	Wake-up Time	Condition
		V _{CC} ^[3]	V _{CCX}			
Read mode (w/ 25ns) ^[1]	I _{CC1} ^[2]	2.19	0.5	mA	N/A	Min. Clock period, duty cycle 100%, VIN = "1/0"
Write mode		0.1	12	mA	N/A	
Erase mode		0.1	12	mA	N/A	
Page Erasure Mode		0.1	12	mA	N/A	
Read mode static current (25-50ns)	I _{CC2}	980	25	μA	N/A	XE=YE=SE="1", between T=T _{acc} and T=50ns, I/O=0mA; later than T=50ns, read mode is turned off, and I/O current is the current of standby mode.
Standby mode	I _{SB}	5.2	20	μA	0	V _{SS} , V _{CCX} , and V _{CC}
Floating mode ^[3]	I _{PD}	0	0	μA	7us	V _{CCX} =0
Typical Value (Room temperature: 25°C)						
Standby mode	I _{SB}	0.4	7.5	μA	0	V _{SS} , V _{CCX} , and V _{CC}
Floating mode ^[3]	I _{PD}	0	0	μA	3.5us	V _{CCX} =0

Note!

- [1] Means the average current, and the peak value is higher than the average one.
- [2] Calculated in different T_{new} clock periods.
 - T_{new} < T_{acc} is not allowed
 - T_{new} = T_{acc}
 - T_{acc} < T_{new} - 50ns: I_{CC1} (new) = (I_{CC1} - I_{CC2})(T_{acc}/T_{new}) + I_{CC2}
 - T_{new} > 50ns: I_{CC1} (new) = (I_{CC1} - I_{CC2})(T_{acc}/T_{new}) + 50ns x I_{CC2}/T_{new} + I_{SB}
 - t > 50ns, I_{CC2} = I_{SB}
- [3] Only supported in user flash in sleep mode.

4.5.2 Timing Parameters^{[1],[5],[6]}

Table 4-19 User Flash Timing Parameters

User Modes	Parameter	Name	Min.	Max.	Unit
Access time ^[2]	WC1	$T_{acc}^{[3]}$	-	25	ns
	TC		-	22	ns
	BC		-	21	ns
	LT		-	21	ns
	WC		-	25	ns
Program/Erase to data storage		T_{nvs}	5	-	μs
Data storage hold time		T_{nvh}	5	-	μs
Data storage hold time (Overall erase)		T_{nvh1}	100	-	μs
Time from data storage to program setup		T_{pgs}	10	-	μs
Program hold time		T_{pgh}	20	-	ns
Program time		T_{prog}	8	16	μs
Write ready time		T_{wpr}	>0	-	ns
Erase hold time		T_{whd}	>0	-	ns
Time from control signal to write/Erase setup		T_{cps}	-10	-	ns
Time from SE to read setup		T_{as}	0.1	-	ns
E pulse high level time		T_{pws}	5	-	ns
Adress/data setup time		T_{ads}	20	-	ns
Adress/data hold time		T_{adh}	20	-	ns
Data hold-up time		T_{dh}	0.5	-	ns
Read mode address hold time ^[3]	WC1	T_{ah}	25	-	ns
	TC		22	-	ns
	BC		21	-	ns
	LT		21	-	ns
	WC		25	-	ns
SE pulse low level time		T_{nws}	2	-	ns
Recovery time		T_{rcv}	10	-	μs
Data storage time		T_{hv}^4	-	6	ms
Erasure time		T_{erase}	100	120	ms
Overall erase time		T_{me}	100	120	ms
Wake-up time from power down to standby mode		T_{wk_pd}	7	-	μs
Standby hold time		T_{sbh}	100	-	ns
V_{CC} setup time		T_{ps}	0	-	ns
V_{CCX} hold time		T_{ph}	0	-	ns

Note!

- [1] The parameter values may change.
- [2] The values are simulation data only.

- [3] After XADR, YADR, XE, and YE are valid, T_{acc} start time is SE rising edge. DOUT is kept until the next valid read operation.
- [4] T_{hv} is the time between write and the next erasure. The same address can not be written twice before erasure, so does the same register. This limitation is for safety.
- [5] Both the rising edge time and falling edge time for all waveform is 1ns.
- [6] TX, YADR, XE, and YE hold time need to be T_{acc} at least, and T_{acc} start from SE rising edge.

4.5.3 Operation Timing Diagrams

Figure 4-1 Read Mode

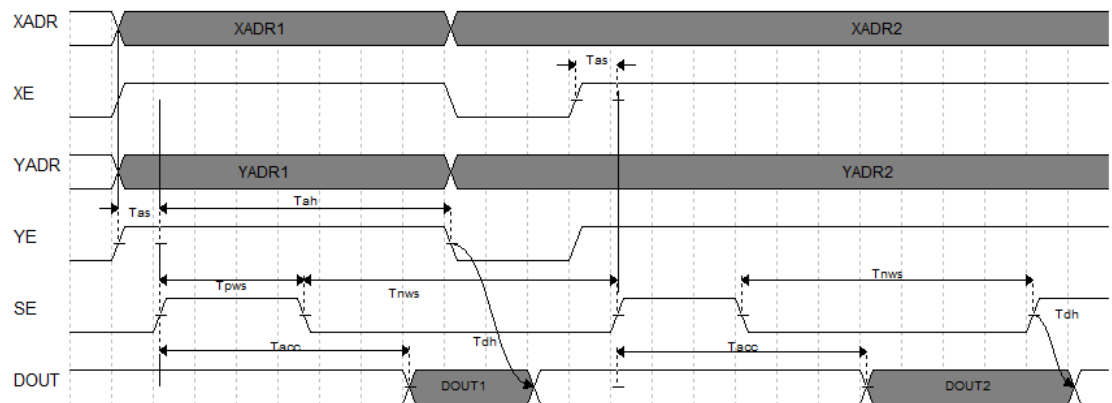


Figure 4-2 Write Mode

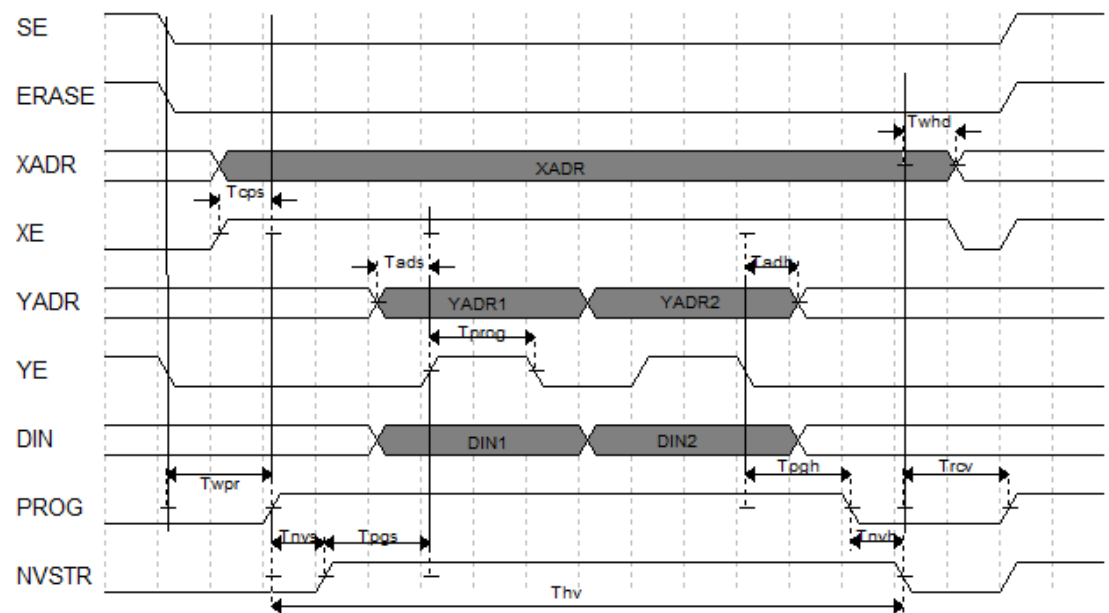
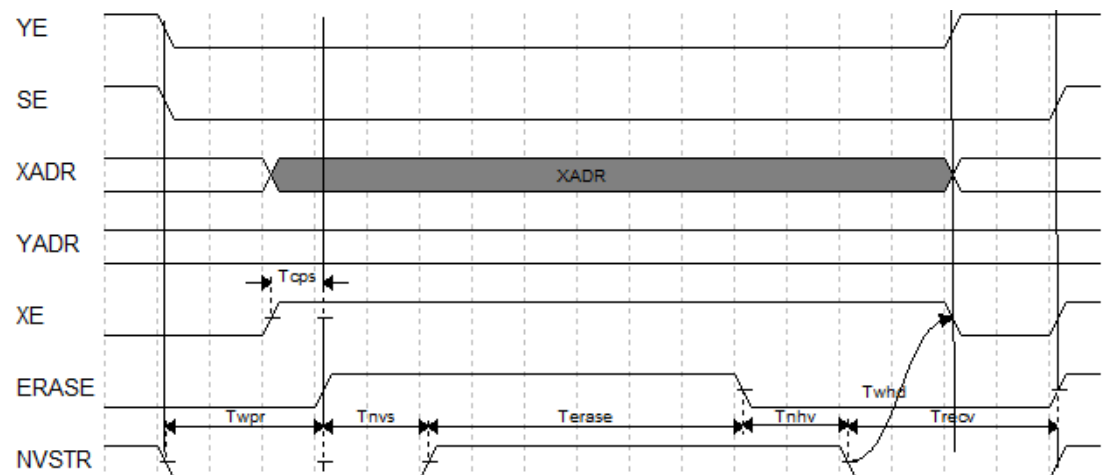


Figure 4-3 Erasure Mode



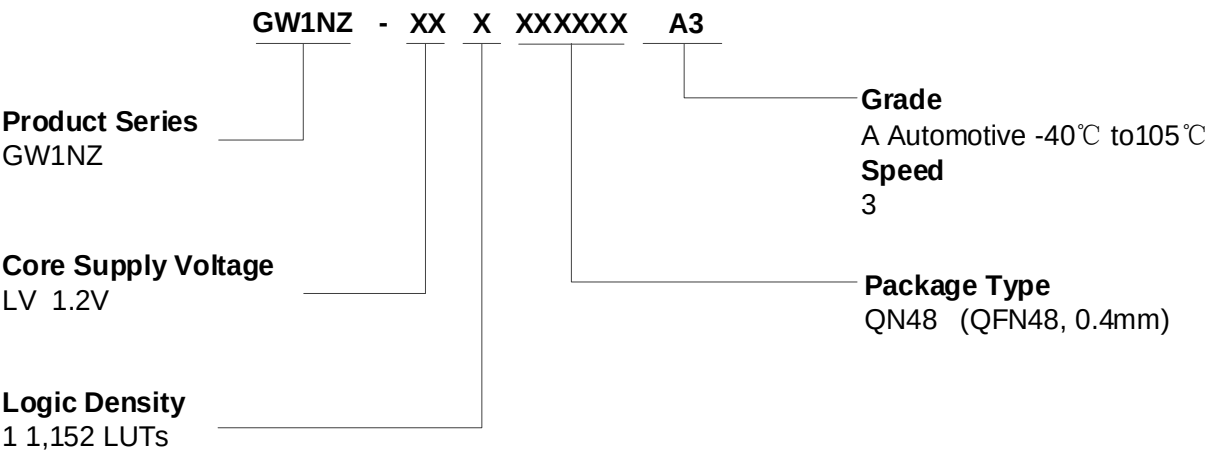
4.6 Configuration Interface Timing Specification

The GW1NZ series of FPGA products (Automotive) GowinCONFIG support six configuration modes: AUTO BOOT, DUAL BOOT, MSPI, SSPI, SERIAL, and CPU. For more detailed information, please refer to [UG290, Gowin FPGA Products Programming and Configuration](#) User Guide.

5Ordering Information

5.1 Part Name

Figure 5-1 Part Naming Example–Production



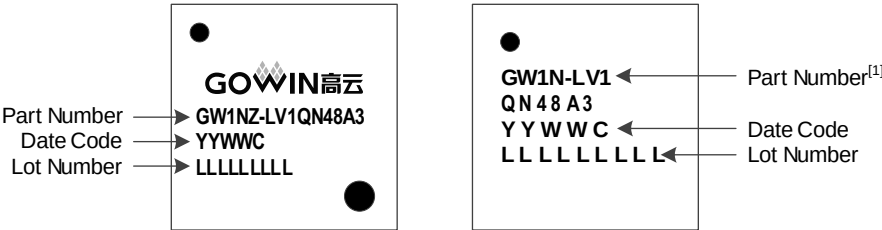
Note!

For the further detailed information about the package type and pin number, please refer to 2.2 Product Resources and 2.3 Package Information.

5.2 Package Mark

The device information of GOWINSEMI is marked on the chip surface, as shown in Figure 5-2.

Figure 5-2 Package Mark



Note!

[1] The first two lines in the right figure above are the “Part Number”.

